

Statistics 516: Practice Final Exam (Solution)

1. The percentage of impurities per batch in a certain type of industrial chemical has the PDF

$$f(x) = kx(1 - x), \quad 0 \leq x \leq 1.$$

A batch having more than 40% impurity cannot be sold. What percentage of batches cannot be sold for this reason?

Solution:

$$1 = \int_0^1 k(x - x^2)dx = k \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 = \frac{1}{6}k.$$
$$\Rightarrow k = 6.$$

Therefore,

$$P(X \geq 0.4) = \int_{0.4}^1 6(x - x^2)dx = (3x^2 - 2x^3) \Big|_{0.4}^1 = 0.648.$$

2. The yield force of a certain type of steel reinforcing bar is normally distributed with a mean of 8,500 lbs. and a standard deviation of 80 lbs. If three bars are used in a certain job, what is the probability that all of them will yield **an average force exceeding 8,700 lbs.?**

Solution: $X \sim N(8500, 80), n = 3. \bar{X} \sim N(8500, \frac{80}{\sqrt{3}}).$

$$P(\bar{X} \geq 8700) = P\left(Z \geq \frac{8700 - 8500}{46.18}\right) = P(Z \geq 4.33) = 0.00000745.$$

3. The grades in a standardized test are distributed normally with mean 70 and variance 81. Top 5% of the students are eligible to apply for a certain program. What is the minimum score required to be eligible to apply for this program?

Solution: $X \sim N(70, 9). P(Z \geq z) = 0.05 \Rightarrow z = 1.645.$

$$x = \mu + z \cdot \sigma = 70 + 1.645 \times 9 = 84.8.$$

4. A uniform random variable has mean 2.75 and variance 0.52. What is the probability that it is less than 3?

Solution: $\mu = \frac{a+b}{2} = 2.75, \sigma^2 = \frac{(a-b)^2}{12} = 0.52. \Rightarrow b = 5.5 - a.$

$$(a - b)^2 = 0.52 \cdot 12 = 6.24 \Rightarrow a - b = 2.498 \approx 2.5.$$

$$a - 5.5 + a = 2.5 \Rightarrow a = 4 \quad \text{and} \quad b = 1.5.$$

$$f(x) = \frac{1}{2.5} = 0.4, \quad 1.5 \leq x \leq 4.$$

$$P(X \leq 3) = 1.5 \times 0.4 = 0.6.$$

5. There are two good bulbs and two bad bulbs in a package. These will be tested one by one in a random order. Find the probability that the second bad bulb is the second bulb tested; the third bulb tested; the fourth bulb tested.

Solution:

$$P(X = 2) = \frac{2}{4} \cdot \frac{1}{3} \cdot 1 = \frac{1}{6}.$$

$$P(X = 3) = \frac{2}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} \cdot 1 + \frac{2}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{3}.$$

$$P(X = 4) = 3 \times \frac{2}{4} \cdot \frac{1}{3} \cdot \frac{2}{2} \cdot \frac{1}{1} = \frac{1}{2}.$$

Thus, $P(X = 2) + P(X = 3) + P(X = 4) = 1$.

6. A fair coin is tossed six times. Given that there are at least three heads, what is the probability that there are exactly four heads?

Solution: Let X be the number of heads, then $X \sim B(6, 0.5)$.

$$P(X = 4 | X \geq 3) = \frac{P(X = 4)}{P(X \geq 3)} = \frac{0.234}{0.656} = 0.357.$$

7. Billy, Silly, and Nilly shoot at a bull's eye. They can hit the bull's eye 70, 80, and 75% of the times, respectively. One of the three is known to have hit the eye. Find the probability that it was Nilly.

Solution: Bayes Rule and conditional probability.

8. The scores of students in a standardized exam are normally distributed with mean 500 and standard deviation 100.

- (a) Jolly scored 654; what is her percentile score?
- (b) Billy scored 540; what is his percentile score?
- (c) Molly is at the 95th percentile of the score distribution; what is Molly's score?

Solution: $X \sim N(500, 100)$

(a) Find $P(X \leq 654)$.

$$P(X \leq 654) = P\left(Z \leq \frac{654 - 500}{100}\right) = 0.9382.$$

Jolly is approximately the 94th percentile.

(b) Find $P(X \leq 540)$.

$$P(X \leq 540) = P\left(Z \leq \frac{540 - 500}{100}\right) = 0.6554.$$

Billy is approximately the 65.5th percentile.

(c) Find $P(X \leq x) = 0.95$.

$$P(Z \leq z = 0.95) \Rightarrow z = 1.645.$$

$$x = \mu + z \cdot \sigma = 500 + 1.645 \cdot 100 = 664.5.$$

Molly's score is 664.5.

9. A normally distributed random variable has mean 40 and the probability that it is between 20 and 60 is 0.5. What is its variance?

Solution: $\mu = 40$, $P(20 \leq X \leq 60) = 0.5$. Find σ^2 .

$$P\left(\frac{20 - 40}{\sigma} \leq Z \leq \frac{60 - 40}{\sigma}\right) = 0.5$$

$$z = 0.674$$

$$\frac{60 - 40}{\sigma} = 0.674 \Rightarrow \sigma = \frac{20}{0.674} = 29.6736 \Rightarrow \sigma^2 = 880.5225.$$